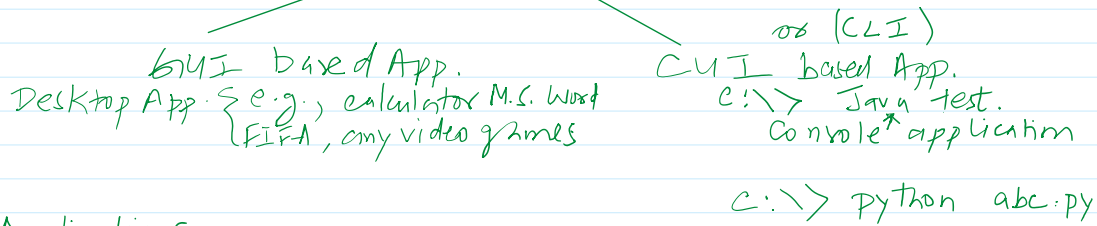


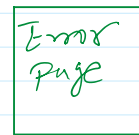
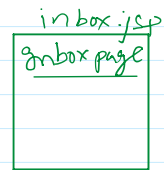
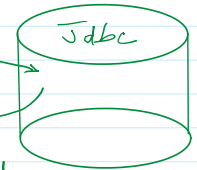
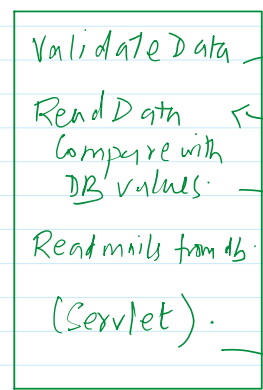
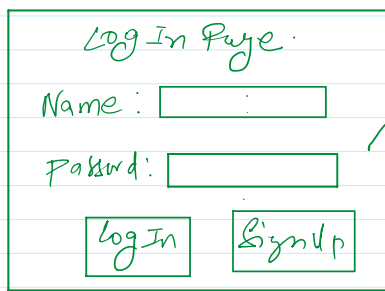
Standalone applications



Web Applications.

gmail
instagram.
github.

- * JDBC
- * Servlets.
- * JSP



View Components -

{ presentation logic }

or JSP (Java Server Pages)

Editions of Java

1. Java Standard Edition. (JDK / JSE / J2SE)

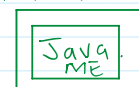
⇒ Core Java + JDBC

② Java Enterprise Edition (J2EE / JEE)

[Servlets + JSP].

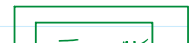
③ Java Micro Edition (J2ME / JME).

Android + SDK.



only for
Android

Kotlin.



cross
platform

New Java Software's:

Java SE

Java on OCI

Oracle WebLogic Server

The world's most popular modern development platform

The world's most popular modern development platform, Java SE is the programming language of choice for enterprise applications. Java SE reduces costs, shortens development time, drives innovation, and

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Java SE

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Java SE

Java on OCI

Oracle WebLogic Server

Java Card

Java Management Service

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[Explore Oracle WebLogic Server](#)

Products

- Oracle WebLogic Server
- Oracle WebLogic Server for Oracle Cloud Infrastructure
- Oracle Coherence
- Helidon



Java Management Service

Documentation

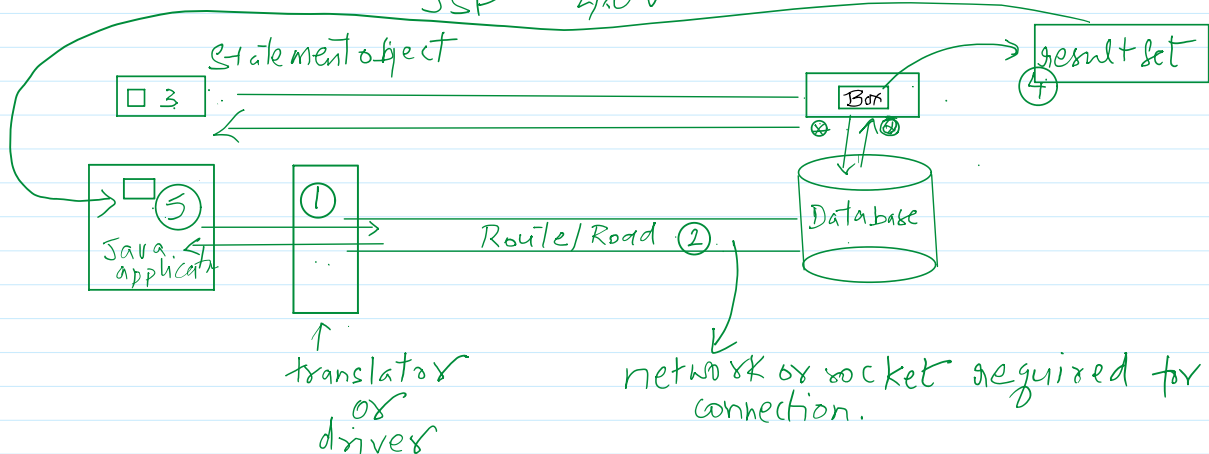
- Interoperable
- Multiapplication, multitenant
- Secure execution environment
- Extensible and updatable



Java Management Service

Learn more about JMS

JSP 4.0v



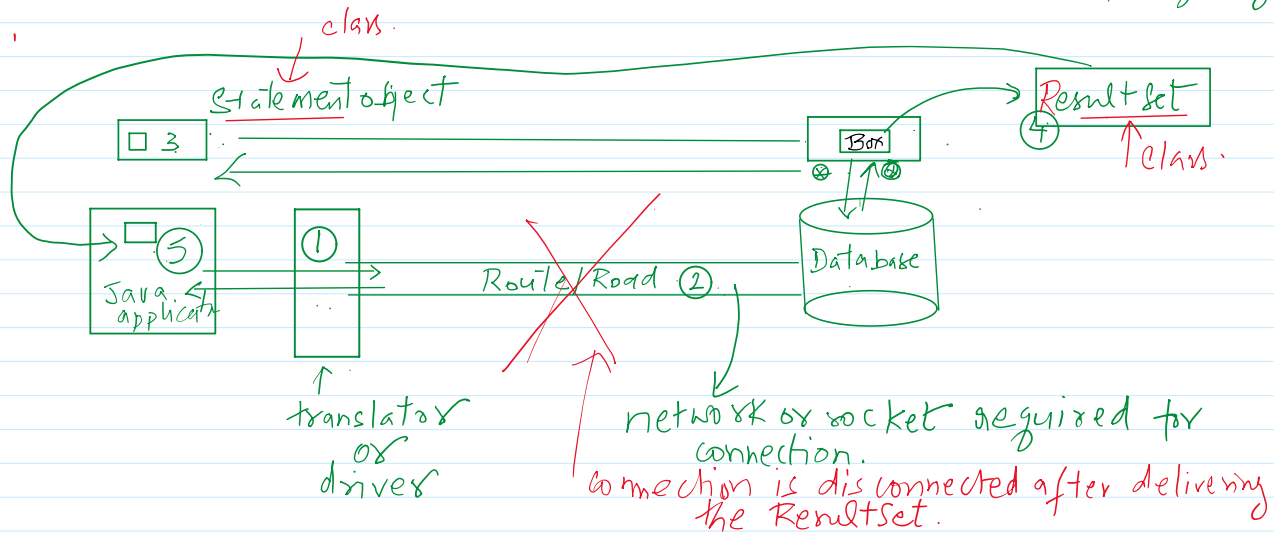
Basic component of JDBC Application:

1. Database Driver : Translator to convert Java specified calls to database specific calls or database specific calls to Java.

2. Network socket or connection :- (Road) or (Route).

3. Statement object (Vehicle) : to send our sql query to database and brings back ResultSet.

4. ResultSet (box) : In which database returns a results of a query



Steps To Develop JDBC Application (Standard Steps).

db (database)

1. Load and Register driver
2. Establish connection b/w Java app. & db
3. Creation of Statement object (Create statement object).
4. Send and execute SQL query.
5. Process results from ResultSet.
6. Close connection.

```
import java.sql.*;
```

```
class JDBC {
```

```
    public static void main (String[] args) {
```

```
        ① → Class.forName("oracle.jdbc.OracleDriver");
```

```
        ② → Connection con = DriverManager.getConnection(url, user, pwd);
```

```
        ③ → Statement st = con.createStatement();
```

```
        ④ → ResultSet rs = st.executeQuery("select * from emp.");
```

```
        ⑤ → while (rs.next()) {
```

```
            System.out.println(rs.getInt(1) + "... " + rs.getString(2) +
```

load the driver
on the
memory.

✓ → write (index, mexi) /

```
S.o.print("In" + rs.getInt(1) + "... " + rs.getString(2) +
        "... " + rs.getDouble(3) + "... " + rs.getString(4));
```

```

        }
    }
}

```

What is job?

It is a technology, which is used to communicate with database.

jdbc is a part of java standard edition. (J2SE/JSE)



JDBC Specification is defined by Java vendor (Oracle).

Who is the implementor of JDBC specification? Database vendor

Database vendor provided implementation is called Driver software.

JDBC Features

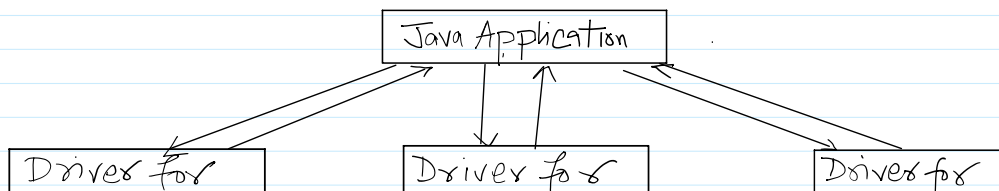
1. Standard API - Database Independent API.
2. Java - Platform Independent technology

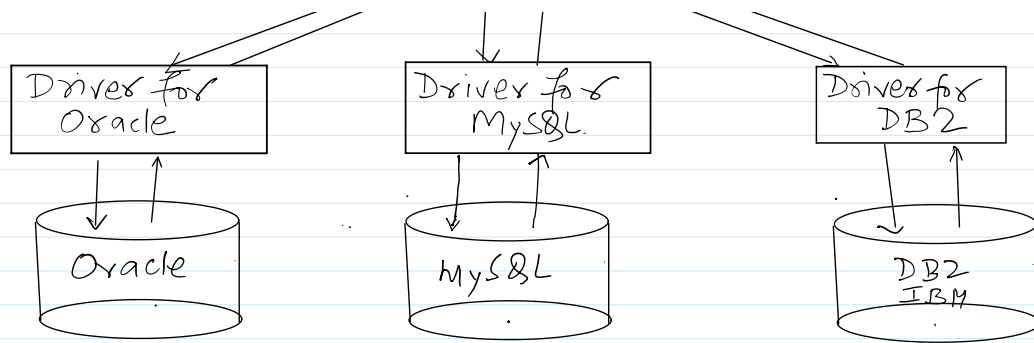
3. **CRUD**

CRUD	Meaning
C	Create
R	Retrieve
U	Update
D	Delete

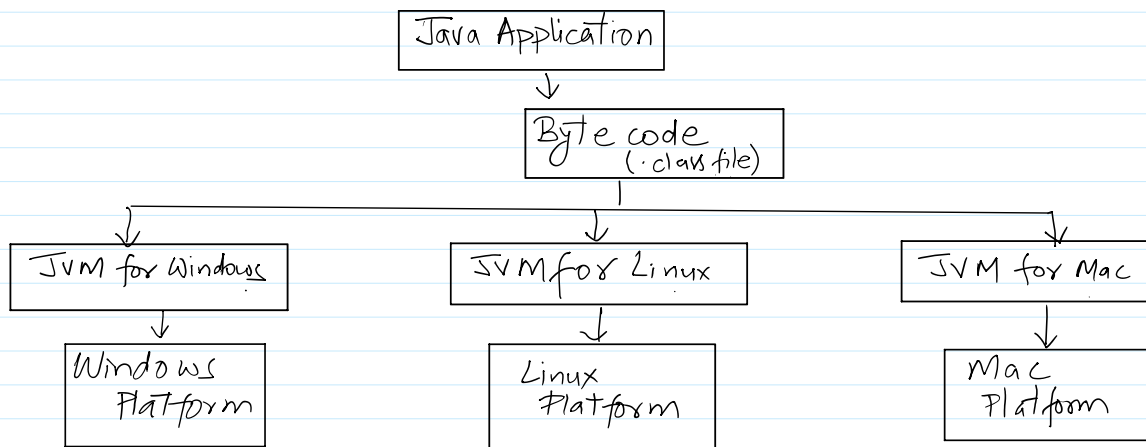
4. Huge vendor support.

How? Java is Platform independent & Database independent





Java is database independent
Driver is database dependent.



Java is platform independent because of **JVM**
But JVM is platform dependent.